
Amoga Sridhar

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Education

University of Illinois, Chicago Chicago, IL
B.S. Biomedical Engineering, Concentration: Neural Engineering May 2028
Awards: Dean's List (Spring '25)

Research Experience and Projects

Neural Imaging Lab (Connectomics), UIC Chicago, IL
Undergraduate Research Assistant Mar 2026 – Present

- Contributing to the startup of a neural imaging lab based on connectomics and electron microscopy of neurons.
- Analyzing neuronal network datasets using WEB Knossos to examine microscale neuronal organization and investigate how local connectivity patterns scale to whole-brain network function.

Low-Cost Spirometer, Biomedical Engineering Society (BMES), UIC Chicago, IL
Design Researcher Aug 2025 – Present

- Researched cost-effective materials like the material used in spirometers.
- Developing CAD designs for our spirometer to improve portability and accommodate different positions.
- Attending general body meetings and participating in discussions on research and industry topics.

Swirl-to-Mix Tenecteplase Device for Blood Clots, Clinical Applications in BME, UIC Chicago, IL
Project Design Engineer Aug 2025 – Dec 2025

- Modified a nasal spray CAD model's geometry and tolerances to meet FDA and ISO requirements for our design.
- Reviewed clinical and engineering considerations of the delivery of Tenecteplase and incorporated findings into a three-page technical report and a one-slide summary, highlighting key engineering decisions and outcomes.

Signal Processing of Pattern Electroretinograms, Modeling Data and Systems, UIC Chicago, IL
Research Analyst Aug 2025 – Dec 2025

- Performed data analysis of 346 data points to compare diseased versus non-diseased groups.
- Contributed to drafting the final project dissertation, interpretation of our results graphically, integrating methodology, and preliminary findings.

Leadership Experience

IEEE Engineering in Medicine and Biology Society (EMBS), UIC Chicago, IL
Marketing Chair May 2026 – Present

- Designing and planning upcoming year-long marketing campaigns using social media, email, and departmental channels, increasing event visibility and member engagement across the UIC BME community.

Asian American Mentor Program (AAMP), UIC Chicago, IL
Peer Mentor & Teaching Fellow Aug 2025 – Dec 2025

- Supported first-year and transfer students by delivery of culturally sensitive topics and presentations, fostering a welcoming and open-discussion environment.

Skills & Interests

- Skills: MATLAB, WEB Knossos, SolidWorks, Arduino IDE, Statistical/Data Analysis, Technical Writing, and Anatomical Dissections
- Interests: Molecular and Physiological Sciences, Information Decision Sciences, and Marketing/Sales Strategies